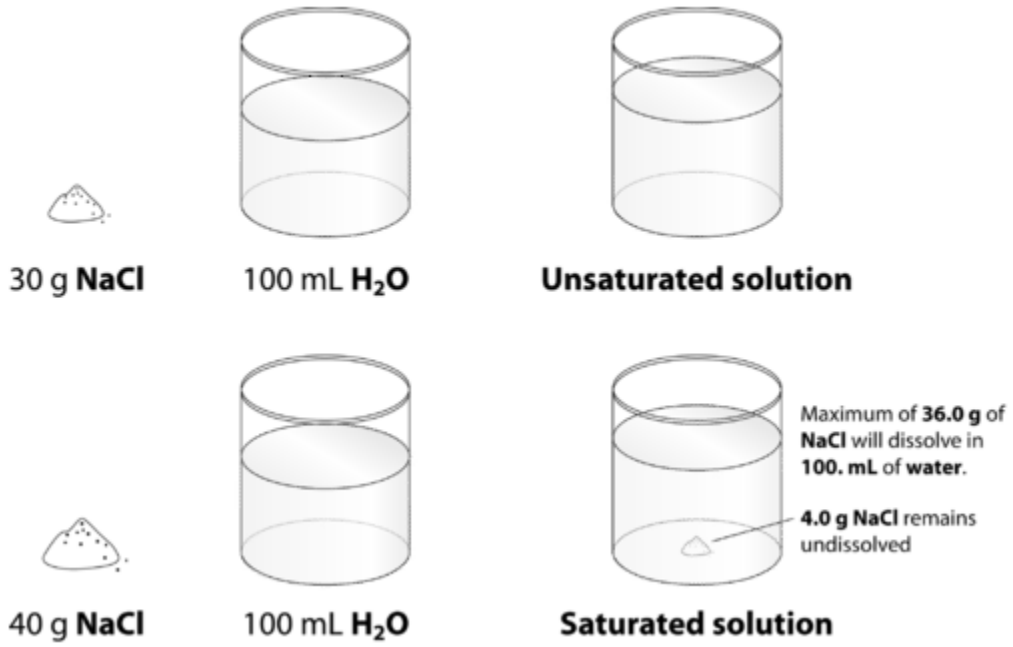
Solubility

Saturated solution: A **saturated solution** is a solution that contains the maximum amount of solute that is capable of dissolving. At $20_{o}C$, the maximum amount of $NaCl$l that will dissolve in \100 of water is $36.0g$. If any more $NaCl$ is added past this point, it will not dissolve, because the solution is saturated. What if more water is added to the solution instead? In that case, more $NaCl$ would be capable of dissolving in the additional solvent.

unsaturated solution is a solution that contains less than the maximum amount of solute that is capable of being dissolved. The figure below illustrates the above process and shows the distinction between unsaturated and saturated.

When $30.0g$30.0g of $NaCl$NaCl is added to $100mL$100mL, it all dissolves, forming an unsaturated



solution. When $40.0g$40.0g is added, $36.0g$36.0g dissolves and $4.0g$4.0g remains undissolved, forming a saturated solution.

How can you tell if a solution is saturated or unsaturated? If more solute is added and it does not dissolve, then the original solution was saturated. If the added solute dissolves, then the original solution was unsaturated. A solution that has been allowed to reach equilibrium, but which has extra undissolved solute at the bottom of the container, must be saturated.

supersaturated solution is a solution that contains more than the maximum amount of solute that is capable of being dissolved at a given temperature. The recrystallization of the excess dissolved solute in a supersaturated solution can be initiated by the addition of a tiny crystal of solute, called a seed crystal. The seed crystal provides a nucleation site on which the excess dissolved crystals can begin to grow. Recrystallization from a supersaturated solution is typically very fast.

Solubility curve

Factors affecting the solubility

Temperature: Most solid solutes become more soluble as temperature increases.

Pressure: Primarily affects gas solubility, where higher pressure leads to increased solubility (Henry's Law).

Polarity: "Like dissolves like" principle - polar solutes dissolve well in polar solvents, and non-polar solutes dissolve well in non-polar solvents.

Particle size: Smaller particles dissolve faster due to increased surface area.

Stirring: Increases the rate of dissolution by facilitating contact between solute and solvent.

Application of solubility

Solubility has many applications in chemistry, medicine, and environmental science.

Chemistry

- **Precipitation**

  The solubility product principle predicts if a precipitate will form when two soluble salts are mixed
- **Water softening**

  The solubility product principle is used to remove calcium and magnesium ions from hard water
- **Preparation of insoluble salts**

  Mixing two soluble salts can form an insoluble salt as a precipitate
- **Control of pH**

  The solubility product principle is used in buffers to maintain a desired pH level
- **Gas solubility**
  Henry's Law describes the relationship between the pressure of a gas and its solubility in a liquid

Medicine

- **Drug delivery**

  Drug solubility is important for controlling drug concentration in the bloodstream and body fluids
- **Solubility curves**
  Solubility curves help pharmacists determine how many drugs dissolve in a given amount of solvent at different temperatures

Environmental science

- **Water treatment**

  Solubility data is used to design facilities that remove hazardous chemicals from drinking water
- **Petroleum products**

- Solubility data is used to remove sulfur-containing gases from petroleum products

Other applications Purification of common salt, Salting out of soap, Manufacture of baking soda, Estimation of silver and barium, and Formulation of artificial blood.